

Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech Degree S3 (R, S) / S3 (PT) (R, S) / S1 (PT) (S, FE) Examination December 2023 (2019 Scheme)

Course Code: EST200**Course Name: Design and Engineering**

Max. Marks: 100

Duration: 3 Hours

PART A*Answer all questions. Each question carries 3 marks*

Marks

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| 1 | Enumerate the two objectives and three constraints in the design of a table fan | (3) |
| 2 | List out three methods to select the best concept from many at the initial stage of design | (3) |
| 3 | Differentiate conventional thinking and creative thinking | (3) |
| 4 | List the techniques to improve the thinking process in a design team | (3) |
| 5 | Write down the name of the product where
(i) aesthetic is essential factor (ii) reliability is most important (iii) aesthetic is not considered | (3) |
| 6 | Compare prototype of a car with its model | (3) |
| 7 | Enumerate six features of a modular design | (3) |
| 8 | Concurrent engineering is better than sequential engineering- Justify | (3) |
| 9 | List the factors affecting the cost of a product | (3) |
| 10 | Enumerate the features of a sustainable product | (3) |

PART B*Answer any one full question from each module. Provide sketches where ever possible. Each question carries 14 marks***Module 1**

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| 11 | You are assigned to design an automatic sanitizer dispenser for your institution. Set the objective and constraints. Apply the standard design procedure to arrive the final concept. | (14) |
| 12 | Explain the method of identifying the customer requirement of an umbrella and the method of finalising the product specification. | (14) |

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Module 2

- 13 Illustrate the design thinking approach for designing a study table for kids within a limited budget. Describe the design thinking process. (14)
- 14 Illustrate three possible conceptual designs of a writing aid for handless person and refine the concepts to obtain the best solution. (14)

Module 3

- 15 Explain the method of developing a mathematical model for a passenger lift. (14)
- 16 Assume that you have completed the design of a new model ceiling fan and a prototype is needed for testing. Communicate your idea to the production department effectively to manufacture the prototype. (14)

Module 4

- 17 Illustrate the modular design of a dining table where three different geometrical varieties can be developed as per the customer requirement. (14)
- 18 Explain the aesthetic, ergonomic and safety considerations incorporated in the design of a baby tricycle. (14)

Module 5

- 19 Describe how to estimate the cost of a chair. Identify the influence of cost on engineering design. Explain the difference in the design of low cost chair and a high quality chair. (14)
- 20 Explain the impact of electronic gadgets on environment and how to minimise the impact with design modification (14)
